Entropie
Relative Entropie von endlichen Quantenzuständen

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Abstract

This is the abstract.

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1 Section 1

1.1 Some Definitions

Question 1.1. Definieren hier die f-Divergenz allgemein für positive lineare Funktionale über von Neumann Algebren. Haben wir dadurch später Vorteile, oder willst du das einfach erstmal so allgemein wie möglich machen? Für die Physik und viele Resultate werden wir schätze ich mal Zustände verwenden.

Definition 1.2. Let $\varphi, \psi : \mathcal{M} \to \mathbb{C}$ be positive lineare functionals over a von Neumann algebra $\mathcal{M}$ such that $\varphi \ll\gg \psi$ was das nochmal genau??? and $f : (0, \infty) \to \mathbb{R}$ be a convex function with $f(1) = 0$ (brauche ich das hier???). The *f-divergence* of φ and ψ is given by

$$\begin{aligned} D_f(\varphi|\psi) &= \int_1^\infty f''(\gamma)E_\gamma(\varphi|\psi) + \gamma^{-3}f''(\gamma^{-1})E_\gamma(\psi|\varphi)d\gamma \\ &= \int_1^\infty f''(\gamma)E_\gamma(\varphi|\psi)d\gamma + \int_0^1 \gamma f''(\gamma)E_{\frac{1}{\gamma}}(\psi|\varphi)d\gamma. \end{aligned} \tag{1}$$

Definition 1.3. Let φ and ψ be as above. (sicher? oder doch Zustände?) The hockeystick divergence $E_\gamma(\varphi|\psi)$ is defined as follows:

$$E_\gamma(\varphi|\psi) := (\psi - \gamma\varphi)_+(\mathbb{1}) \tag{2}$$

Some facts about the hockeystick divergence.

Proposition 1.4. Let φ_1, φ_2 be positive lineare functionals over a von Neumann algebra, and p_1, p_2 are (pos. lin. functionals or states?)

$$\lim_{t\to\infty} E_t(\varphi_1|\varphi_2) = \varphi_1(p_2^\perp). \tag{3}$$

$$\lim_{t\to\infty} E_t(\varphi_1, \varphi_2) = 0 \iff p_1 \leq p_2. \tag{4}$$

1.2 First Inequalitie

In [SV16] there are a lot of inequalities for f-divergences in the context of probability measures.

We want to generalize them for states over arbitrary von Neumann algebras (even vNA of type III).

We start with inequalitie (84) from [SV16].

Question 1.5. Wann ist $D_f(\varphi|\psi)$ endlich? Evtl wenn ψ normal ist?

Proposition 1.6. Let $f, g : (0, \infty) \to \mathbb{R}$ convex functions with $f(1) = g(1)$, $f'(1) = g'(1)$ and $f(t) \leq g(t)$ for all $t \in (0, \infty)$. Also let φ, ψ be positive lineare functionals over a von Neumann algebra such that $\varphi \ll\gg \psi$ $(D_f(\varphi|\psi)), D_g(\varphi|\psi)$ are finite

$$D_f(\varphi|\psi) \leq D_g(\varphi|\psi). \tag{5}$$

Remark. Alexandrovs theorem states that every convex function defined on an open domain is twice differentiable almost everywhere.

Remark. By assumption $f'(1) = g'(1)$ indirectly require, that f and g are differentiable at $x = 1$. Therefore, we cannot put in e.g. the *total variational distance* (see [SV16] eq. (55)).

Proof. We begin by reformulating the f-divergence given in definition 1.2.

$$\begin{aligned}
D_f(\varphi|\psi) &= \int_1^\infty f''(\gamma)E_\gamma(\varphi|\psi) + \gamma^{-3}f''(\gamma^{-1})E_\gamma(\psi|\varphi)d\gamma \\
&= \int_1^\infty f''(\gamma)E_\gamma(\varphi|\psi)d\gamma + \int_0^1 \gamma f''(\gamma)E_{\frac{1}{\gamma}}(\psi|\varphi)d\gamma \\
&\overset{(a)}{=} \int_1^\infty f''(\gamma)E_\gamma(\varphi|\psi)d\gamma + \int_0^1 \gamma f''(\gamma)(\frac{1}{\gamma}E_\gamma(\varphi|\psi) + \psi(\mathbb{1}) - \frac{1}{\gamma}\varphi(\mathbb{1}))d\gamma \\
&= \int_1^\infty f''(\gamma)E_\gamma(\varphi|\psi)d\gamma + \int_0^1 f''(\gamma)E_\gamma(\varphi|\psi)d\gamma + \int_0^1 f''(\gamma)(\gamma\psi(\mathbb{1}) - \varphi(\mathbb{1}))d\gamma \\
&= \int_0^\infty f''(\gamma)E_\gamma(\varphi|\psi)d\gamma + \psi(\mathbb{1})\int_0^1 \gamma f''(\gamma)d\gamma - \varphi(\mathbb{1})\int_0^1 f''(\gamma)d\gamma \\
&\overset{(b)}{=} [f'(\gamma)E_\gamma(\varphi|\psi)]_0^\infty - \int_0^\infty f'(\gamma)E'_\gamma(\varphi|\psi)d\gamma + \psi(\mathbb{1})\left([\gamma f'(\gamma)]_0^1 - \int_0^1 f'(\gamma)d\gamma\right) - \varphi(\mathbb{1})[f'(\gamma)]_0^1 \\
&\overset{(c)}{=} [f'(\gamma)E_\gamma(\varphi|\psi)]_0^\infty - \left([f(\gamma)E'_\gamma(\varphi|\psi)]_0^\infty - \int_0^\infty f(\gamma)E''_\gamma(\varphi|\psi)d\gamma\right) \\
&\quad + \psi(\mathbb{1})\left([\gamma f'(\gamma)]_0^1 - [f(\gamma)]_0^1\right) - \varphi(\mathbb{1})[f'(\gamma)]_0^1 \\
&= \lim_{\gamma\to\infty} f'(\gamma)E_\gamma(\varphi|\psi) - \lim_{\gamma\to 0} f'(\gamma)E_\gamma(\varphi|\psi) - \left(\lim_{\gamma\to\infty} f(\gamma)E'_\gamma(\varphi|\psi) - \lim_{\gamma\to 0} f(\gamma)E'_\gamma(\varphi|\psi)\right) \\
&\quad + \int_0^\infty f(\gamma)E''_\gamma(\varphi|\psi)d\gamma + \psi(\mathbb{1})\left(1f'(1) - \lim_{\gamma\to 0}\gamma f'(\gamma) - f(1) + \lim_{\gamma\to 0} f(\gamma)\right) \\
&\quad - \varphi(\mathbb{1})\left(f'(1) - \lim_{\gamma\to 0} f'(\gamma)\right)
\end{aligned} \tag{6}$$

(a) state exchange, □

2 Section 2

2.1 Subsection A

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References

[SV16] Igal Sason and Sergio Verdu. "f -Divergence Inequalities". In: *IEEE Transactions on Information Theory* 62.11 (Nov. 2016), pp. 5973–6006. ISSN: 1557-9654. DOI: 10.1109/tit.2016.2603151. URL: http://dx.doi.org/10.1109/TIT.2016.2603151.